



# The Wave Equation

## Definition

The wave equation is given by  $u_{tt} = c^2 u_{xx}$ , or equivalently

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2},$$

where  $c$  is the **wave speed**

Describes a wave traveling in the  $x$  direction.

Note that the wave equation is a hyperbolic equation, since  $B^2 - 4AC = 4c^2 > 0$ .



# The Wave Equation

The full wave equation must be specified with an initial condition and boundary conditions:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

$$u(x, 0) = f(x), \quad x \in [a, b] \quad \text{initial profile}$$

$$u_t(x, 0) = g(x), \quad x \in [a, b] \quad \text{initial derivative}$$

$$u(a, t) = l(t), \quad t \in [t_0, T] \quad \text{boundary on the left boundary}$$

$$u(b, t) = r(t), \quad t \in [t_0, T] \quad \text{boundary on the right boundary}$$

Unlike the heat equation, additional information is required to specify the initial derivative.

## (Forward) Finite Difference Method

We approximately solve the wave equation  $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$  as

$$\frac{u_{i,j+1} - 2u_{i,j} + u_{i,j-1}}{k^2} + \mathcal{O}(k^2) = c^2 \left[ \frac{u_{i+1,j} - 2u_{i,j} + u_{i-1,j}}{h^2} + \mathcal{O}(h^2) \right]$$

Note that the local (or step) error is  $\mathcal{O}(h^2) + \mathcal{O}(k^2)$ .

Letting  $\sigma = ck/h$ , this system yields:

$$u_{i,j+1} = \sigma^2 u_{i+1,j} + (2 - 2\sigma^2) u_{i,j} + \sigma^2 u_{i-1,j} - u_{i,j-1}$$

(See notes.)

## (Forward) Finite Difference Method

With  $\mathbf{u}_j = [u_{1,j}, u_{2,j}, \dots, u_{n-1,j}]^T$  and  $\mathbf{s}_j = [l(t_j), 0, \dots, 0, r(t_j)]^T$ , the finite difference equation takes the form

$$\mathbf{u}_{j+1} = B\mathbf{u}_j - \mathbf{u}_{j-1} + \sigma^2 \mathbf{s}_j,$$

where

$$B = \begin{bmatrix} 2 - 2\sigma^2 & \sigma^2 & 0 & \dots & 0 \\ \sigma^2 & 2 - 2\sigma^2 & \sigma^2 & \ddots & \vdots \\ 0 & \sigma^2 & 2 - 2\sigma^2 & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & \sigma^2 \\ 0 & \dots & 0 & \sigma^2 & 2 - 2\sigma^2 \end{bmatrix}.$$

(See notes for derivation. Also, observe that  $B$  is identical to  $I + A$  from before except now we have  $\sigma \mapsto \sigma^2$ .)



## (Forward) Finite Difference Method

To compute the values  $\mathbf{u}_{i,1}$  at the first time step  $t_1$ , we use the derivative initial condition to obtain

$$\mathbf{u}_{j+1} = \frac{1}{2}B\mathbf{u}_j + k\mathbf{g} + \frac{1}{2}\sigma^2\mathbf{s}_j,$$

where  $\mathbf{g} = [g(x_1), \dots, g(x_{n-1})]^T$ . For the remaining time steps we use

$$\mathbf{u}_{j+1} = B\mathbf{u}_j - \mathbf{u}_{j-1} + \sigma^2\mathbf{s}_j.$$

(See notes.)



# CFL Condition

## Theorem (Courant-Friedrichs-Lewy)

The iteration

$$\mathbf{u}_{j+1} = B\mathbf{u}_j - \mathbf{u}_{j-1} + \sigma^2 \mathbf{s}_j.$$

with wave speed  $c > 0$  is stable if the **CFL number** satisfies

$$\sigma = ck/h \leq 1.$$

(See notes for proof.)